

## SAFETY DATA SHEET

Creation Date 29-Jul-2011

Revision Date 06-Jan-2023

Revision Number 6

### 1. Identification

**Product Name** tert-Butyllithium, 1.6-3.2 M (15-30%) solution in Heptane

**Cat No. :** AC445920000; AC445921000

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                                        |              |
|------------------------------------------------------------------------|--------------|
| Flammable liquids                                                      | Category 1   |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1   |
| Pyrophoric liquids                                                     | Category 1   |
| Skin Corrosion/Irritation                                              | Category 1 B |
| Serious Eye Damage/Eye Irritation                                      | Category 1   |
| Specific target organ toxicity (single exposure)                       | Category 3   |
| Target Organs - Respiratory system, Central nervous system (CNS).      |              |
| Specific target organ toxicity - (repeated exposure)                   | Category 1   |
| Target Organs - Liver.                                                 |              |
| Aspiration Toxicity                                                    | Category 1   |

#### Label Elements

##### **Signal Word**

Danger

### **Hazard Statements**

Extremely flammable liquid and vapor  
In contact with water releases flammable gases which may ignite spontaneously  
Catches fire spontaneously if exposed to air  
May be fatal if swallowed and enters airways  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May cause drowsiness or dizziness  
Causes damage to organs through prolonged or repeated exposure



### **Precautionary Statements**

#### **Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Do not allow contact with air  
Keep away from any possible contact with water, because of violent reaction and possible flash fire  
Handle under inert gas. Protect from moisture  
Keep cool

#### **Response**

Immediately call a POISON CENTER or doctor/physician

#### **Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

#### **Skin**

Wash contaminated clothing before reuse  
IF ON SKIN: Immerse in cool water/wrap with wet bandages  
Brush off loose particles from skin. Immerse in cool water/wrap with wet bandages

#### **Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

#### **Ingestion**

Do NOT induce vomiting  
Rinse mouth

#### **Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

#### **Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store contents under inert gas  
Store in a dry place. Store in a closed container

#### **Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects  
Reacts violently with water

### 3. Composition/Information on Ingredients

| Component                     | CAS No   | Weight % |
|-------------------------------|----------|----------|
| n-Heptane                     | 142-82-5 | 70-85    |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | 15-30    |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. Risk of serious damage to the lungs (by aspiration). If not breathing, give artificial respiration.                                                                                                                                                |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### 5. Fire-fighting measures

|                                         |                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Powder. Limestone powder. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | Carbon dioxide (CO <sub>2</sub> ), DO NOT USE WATER                         |
| <b>Flash Point</b>                      | -4 °C                                                                       |
| <b>Method -</b>                         | No information available                                                    |
| <b>Autoignition Temperature</b>         | No information available                                                    |
| <b>Explosion Limits</b>                 |                                                                             |
| <b>Upper</b>                            | 7.40 vol %                                                                  |
| <b>Lower</b>                            | 1.20 vol %                                                                  |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                    |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                    |

#### Specific Hazards Arising from the Chemical

Extremely flammable. Spontaneously flammable in air. Reacts violently with water. Contact with water liberates extremely flammable gases. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air. Do not allow run-off from fire-fighting to

enter drains or water courses.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

#### NFPA

Health  
3

Flammability  
4

Instability  
2

Physical hazards  
W

### 6. Accidental release measures

#### Personal Precautions

Use personal protective equipment as required. Remove all sources of ignition. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Take precautionary measures against static discharges. Do not get in eyes, on skin, or on clothing.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

#### Methods for Containment and Clean Up

Remove all sources of ignition. Do not expose spill to water. Soak up with inert absorbent material. Take precautionary measures against static discharges. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment.

### 7. Handling and storage

#### Handling

Use only under a chemical fume hood. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not allow contact with water. Do not get water inside containers. Handle under an inert atmosphere. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

#### Storage.

Keep away from heat, sparks and flame. Keep under nitrogen. Keep refrigerated. Flammables area. Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Incompatible Materials. Strong oxidizing agents. Halogens. Acids. Alcohols.

### 8. Exposure controls / personal protection

#### Exposure Guidelines

| Component | ACGIH TLV                     | OSHA PEL                                                                                                                                                                            | NIOSH                                                                                                             | Mexico OEL (TWA)              |
|-----------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------|
| n-Heptane | TWA: 400 ppm<br>STEL: 500 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 1600 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 2000 mg/m <sup>3</sup><br>TWA: 500 ppm<br>TWA: 2000 mg/m <sup>3</sup> | IDLH: 750 ppm<br>TWA: 85 ppm<br>TWA: 350 mg/m <sup>3</sup><br>Ceiling: 440 ppm<br>Ceiling: 1800 mg/m <sup>3</sup> | TWA: 400 ppm<br>STEL: 500 ppm |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** low boiling organic solvent. Type AX. Brown. conforming to EN371. or. Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                  |
|----------------------------------------|----------------------------------|
| Physical State                         | Liquid                           |
| Appearance                             | Yellow                           |
| Odor                                   | Petroleum distillates            |
| Odor Threshold                         | No information available         |
| pH                                     | No information available         |
| Melting Point/Range                    | No data available                |
| Boiling Point/Range                    | No information available         |
| Flash Point                            | -4 °C                            |
| Evaporation Rate                       | No information available         |
| Flammability (solid,gas)               | Not applicable                   |
| Flammability or explosive limits       |                                  |
| Upper                                  | 7.40 vol %                       |
| Lower                                  | 1.20 vol %                       |
| Vapor Pressure                         | No information available         |
| Vapor Density                          | No information available         |
| Specific Gravity                       | 0.690                            |
| Solubility                             | No information available         |
| Partition coefficient; n-octanol/water | No data available                |
| Autoignition Temperature               | No information available         |
| Decomposition Temperature              | No information available         |
| Viscosity                              | No information available         |
| Molecular Formula                      | C <sub>4</sub> H <sub>9</sub> Li |
| Molecular Weight                       | 64.06                            |

## 10. Stability and reactivity

|                               |                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>        | Yes                                                                                                                                                                |
| <b>Stability</b>              | Pyrophoric: Spontaneously flammable in air. Reacts violently with water. Air sensitive. Moisture sensitive.                                                        |
| <b>Conditions to Avoid</b>    | Incompatible products. Heat, flames and sparks. Exposure to moist air or water. Exposure to air. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b> | Strong oxidizing agents, Halogens, Acids, Alcohols                                                                                                                 |

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** Contact with water liberates extremely flammable gases. Pyrophoric: Spontaneously flammable in air.

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component | LD50 Oral         | LD50 Dermal                  | LC50 Inhalation              |
|-----------|-------------------|------------------------------|------------------------------|
| n-Heptane | >2000 mg/kg (rat) | LD50 = 3000 mg/kg ( Rabbit ) | LC50 > 73.5 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                        | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------|----------|------------|------------|------------|------------|------------|
| n-Heptane                        | 142-82-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Lithium,<br>(1,1-dimethylethyl)- | 594-19-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)

**STOT - repeated exposure** Liver

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Algae | Freshwater Fish                           | Microtox   | Water Flea         |
|-----------|------------------|-------------------------------------------|------------|--------------------|
| n-Heptane | Not listed       | LC50: = 375.0 mg/L, 96h<br>(Cichlid fish) | Not listed | EC50: >10 mg/L/24h |

**Persistence and Degradability** May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** .

| Component | log Pow |
|-----------|---------|
| n-Heptane | 4.66    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN3394  
**Proper Shipping Name** ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE  
**Hazard Class** 4.2  
**Subsidiary Hazard Class** 4.3  
**Packing Group** I

### TDG

UN-No UN3394  
**Proper Shipping Name** ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE  
**Hazard Class** 4.2  
**Subsidiary Hazard Class** 4.3  
**Packing Group** I

### IATA

FORBIDDEN FOR IATA TRANSPORT  
 UN-No UN3394  
**Proper Shipping Name** ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE.  
 FORBIDDEN FOR IATA TRANSPORT  
**Hazard Class** 4.2  
**Subsidiary Hazard Class** 4.3  
**Packing Group** I

### IMDG/IMO

UN-No UN3394  
**Proper Shipping Name** ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE  
**Hazard Class** 4.2  
**Subsidiary Hazard Class** 4.3  
**Packing Group** I

## 15. Regulatory information

### United States of America Inventory

| Component                     | CAS No   | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|-------------------------------|----------|------|--------------------------------------------------|--------------------------------|
| n-Heptane                     | 142-82-5 | X    | ACTIVE                                           | -                              |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | X    | ACTIVE                                           | -                              |

Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)** Not applicable

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                     | CAS No   | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL        |
|-------------------------------|----------|-----|------|-----------|-------|------|------|------|-------|-------------|
| n-Heptane                     | 142-82-5 | X   | -    | 205-563-8 | X     | X    | X    | X    | X     | KE-18271    |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | -   | X    | 209-831-5 | X     | X    | X    | X    | X     | 2014-3-6117 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA** - Occupational Safety and Health Administration Not applicable

**CERCLA** Not applicable

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

### **U.S. State Right-to-Know Regulations**

| Component | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------|---------------|------------|--------------|----------|--------------|
| n-Heptane | X             | X          | X            | -        | X            |

### **U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant Y

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

### **Authorisation/Restrictions according to EU REACH**

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances | REACH (1907/2006) - Annex XVII - Restrictions | REACH Regulation (EC 1907/2006) article 59 - |
|-----------|--------|--------------------------------------------|-----------------------------------------------|----------------------------------------------|
|-----------|--------|--------------------------------------------|-----------------------------------------------|----------------------------------------------|



|                               |          | Subject to Authorization | on Certain Dangerous Substances                                    | Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------|----------|--------------------------|--------------------------------------------------------------------|----------------------------------------------------------|
| n-Heptane                     | 142-82-5 | -                        | Use restricted. See item 75.<br>(see link for restriction details) | -                                                        |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | -                        | -                                                                  | -                                                        |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                     | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| n-Heptane                     | 142-82-5 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                     | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| n-Heptane                     | 142-82-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

**Creation Date** 29-Jul-2011  
**Revision Date** 06-Jan-2023  
**Print Date** 06-Jan-2023

**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**